Cart lab: change driving force

antocciphysics | TEACHER KEY | CPA

Learning target: Use controlled data to describe how net force affects acceleration.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 9; 3 course days later.*
The straight-moving rover can have zero net force. The turning rover changes velocity direction and needs net force.

02. Explain a worked example (worked)
Plot F against a: (1, 2),(3, 6); slope 2 kg, the mass.

03. Read a proportional pattern (model)
It doubles; at 8 N, a=4 m/s².

04. Keep mass constant (practice)
Driving weight and total moving mass change. Transfer mass from cart to hanger so total stays constant.

05. Graph the pattern (reason)
Line through (2, 1),(4, 2),(6, 3); ideally through origin. Axes force N anda m/s².

06. Choose a method (mixed) *Review: Unit 1, Day 11; 1 course days later.*
5 m/s²; the mass doubles at the same net force. Keep the net force fixed, including resistance.

07. Exit check (exit)
3 m/s² because force is 1.5 times as large at same mass.

08. STRETCH: Apply it another way (stretch)
3, 6, 9 N; total moving mass and relevant conditions fixed.

Teaching note: Lab companion. Preserve total moving mass and use teacher-approved releases. Keep supplied model data distinct from measurements.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 9; 3 course days later.*
Tangent velocity direction changes, so acceleration and net force are inward and nonzero. Constant speed is insufficient for equilibrium.

02. Explain a worked example (worked)
Plot F against a: (1, 2),(3, 6); slope 2 kg, the mass.

03. Interpret a force graph (model)
Slope 1(m/s²)/N=1/kg; mass 1 kg.

04. Design the controlled trial (practice)
Same parts remain in system; their location changes. Table m_h, F_{drive}, M, 3 a trials, mean. Keep release/track consistent; record friction effects.

05. Both variables change (reason)
aA=2 m/s²; aB=3 m/s²; ratio 1.5. Mass also doubles.

06. Choose a method (mixed) *Review: Unit 1, Day 11; 1 course days later.*
Net force 0.15 N; new a≈0.2143 m/s². Extra resistance reduces net force and acceleration if driving force is unchanged.

07. Exit check (exit)
Mass .50 kg; a=3 m/s². Resistance subtracts from driving force, so acceleration may not pass through origin on driving force plot.

08. STRETCH: Apply it another way (stretch)
F_{pull}=2 a+1, so resistance 1 N. Intercept at a=0 is 1 N; slope mass 2 kg.

Teaching note: Lab companion. Preserve total moving mass and use teacher-approved releases. Keep supplied model data distinct from measurements.